

Deep Learning

Introduction to Deep Learning

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks (ANNs) with multiple layers (deep architectures) to model complex patterns in data. It is widely used in image recognition, natural language processing, speech recognition, and more.

Key Features:

- Learns automatically from data
 - Handles large and complex datasets
 - Uses multiple layers to extract features
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Types of Deep Learning

1. Supervised Learning

- Uses labeled data
- Example: Image classification, spam detection

2. Unsupervised Learning

- Uses unlabeled data
- Example: Clustering, anomaly detection

3. Reinforcement Learning

- Learns through rewards and penalties
 - Example: Game playing (AlphaGo), robotics
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I .Artificial Neural Network (ANN)

ANN is a computational model inspired by the human brain. It consists of layers of interconnected nodes (neurons).

- Input Layer: Receives input data
- Hidden Layers: Perform computations
- Output Layer: Produces final result

Working of ANN

1. Input data is fed into the network
 2. Weights and biases are applied
 3. Activation function is used
 4. Output is generated
 5. Loss is calculated
 6. Backpropagation updates weights
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II .Convolutional Neural Network (CNN)

CNN is mainly used for image processing and computer vision tasks.

- Convolution Layer: Extracts features
- Pooling Layer: Reduces dimensions
- Fully Connected Layer: Final classification

Applications:

- Image classification
 - Face detection
 - Medical image analysis
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III. Recurrent Neural Network (RNN)

RNN is used for sequential data where previous information is important.

- Has memory
- Processes sequences step-by-step

Applications:

- Text generation
 - Language translation
 - Speech recognition
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